Started on	Monday, 18 May 2026, 9:41 AM
State	Finished
Completed on	Monday, 18 May 2026, 10:12 AM
Time taken	31 mins 49 secs
Grade	**87.00** out of 100.00

Question 1

Correct

Mark 1.00 out of 1.00

Which of the following is correct? Please select all that apply.

Select one or more:

- A. String[] list = {"red", "yellow", "green"}; ✔ See the note before CodeAnimation TestArrayArguments.
- B. String list = {"red", "yellow", "green"};
- C. String[] list = new String{"red", "yellow", "green"};
- D. String list = new String{"red", "yellow", "green"};
- E. String[] list = new String[]{"red", "yellow", "green"}; ✔ See the note before CodeAnimation TestArrayArguments.

See the note before CodeAnimation TestArrayArguments.

See the note before CodeAnimation TestArrayArguments.

The correct answers are: String[] list = new String[]{"red", "yellow", "green"};, String[] list = {"red", "yellow", "green"};

Question 2

Correct

Mark 2.00 out of 2.00

Suppose your method does not return any value, which of the following keywords can be used as a return type?

Select one:

- A. void ✔ See the third paragraph in this section.
- B. int
- C. double
- D. public
- E. None of these

See the third paragraph in this section.

See the third paragraph in this section.

The correct answer is: void

Question 3

Correct

Mark 1.00 out of 1.00

How can you initialize an array of two characters to 'a' and 'b'? Please select all that apply.

Select one or more:

- A. char[] charArray = new char[2]; charArray = {'a', 'b'}; ✖
- B. char[2] charArray = {'a', 'b'};
- C. char[] charArray = {'a', 'b'}; ✔ See Section 7.2.5.
- D. char[] charArray = new char[2]; charArray[0] = 'a'; charArray[1] = 'b'; ✔ See Section 7.2.5.

See Section 7.2.5.

See Section 7.2.5.

The correct answers are: char[] charArray = {'a', 'b'};, char[] charArray = new char[2]; charArray[0] = 'a'; charArray[1] = 'b';

Question 4

Correct

Mark 2.00 out of 2.00

Which of the following is the best for generating random integer 0 or 1?

Select one:

- A. (int)Math.random()
- B. (int)Math.random() + 1
- C. (int)(Math.random() + 0.5) ✔ 50% of (Math.random() + 0.5) is <= 1 and 50% of (Math.random() + 0.5) is > 1. So, C is the correct answer.
- D. (int)(Math.random() + 0.2)
- E. (int)(Math.random() + 0.8)

50% of (Math.random() + 0.5) is <= 1 and 50% of (Math.random() + 0.5) is > 1. So, C is the correct answer.

50% of (Math.random() + 0.5) is <= 1 and 50% of (Math.random() + 0.5) is > 1. So, C is the correct answer.

The correct answer is: (int)(Math.random() + 0.5)

Question 5

Correct

Mark 2.00 out of 2.00

Consider the following incomplete code. The missing method body should be _____.

```
public class Test {
  public static void main(String[] args) {
    System.out.println(f(5));
  }

  public static int f(int number) {
    // Missing body
  }
}
```

Select one:

- A. return "number";
- B. System.out.println(number);
- C. System.out.println("number");
- D. return number; ✔ The method returns and int. So D is correct.

The method returns and int. So D is correct.

The method returns and int. So D is correct.

The correct answer is: return number;

Question 6

Partially correct

Mark 0.50 out of 1.00

Which of the following declarations are correct? Please select all that apply.

Select one or more:

- A. public static double... print(double d1, double d2)
- B. public static void print(int n, double... numbers)
- C. public static void print(String... strings, double... numbers) ✖
- D. public static void print(double... numbers, String name)
- E. public static void print(double... numbers) ✔ Only one variable-length parameter may be specified in a method and this parameter must be the last parameter. The method return type cannot be a variable-length parameter.

Only one variable-length parameter may be specified in a method and this parameter must be the last parameter. The method return type cannot be a variable-length parameter.

The correct answers are: public static void print(double... numbers), public static void print(int n, double... numbers)

Question 7

Correct

Mark 2.00 out of 2.00

Each time a method is invoked, the system stores parameters and local variables in an area of memory, known as _____, which stores elements in last-in first-out fashion.

Select one:

- A. storage area
- B. a heap
- C. an array
- D. a stack ✔ See the discussion on the activation record.

See the discussion on the activation record.

See the discussion on the activation record.

The correct answer is: a stack

Question 8

Correct

Mark 1.00 out of 1.00

If a key is not in the list, the binarySearch method returns _____.

Select one:

- A. -(insertion point + 1) ✔ See Listing 7.7.
- B. insertion point
- C. -insertion point
- D. insertion point - 1

See Listing 7.7.

See Listing 7.7.

The correct answer is: -(insertion point + 1)

Question 9

Correct

Mark 1.00 out of 1.00

Analyze the following code.

```
int[] list = new int[5];
list = new int[6];
```

Select one:

- A. The code has runtime errors because the variable list cannot be changed once it is assigned.
- B. The code has compile errors because you cannot assign a different size array to list.
- C. The code has compile errors because the variable list cannot be changed once it is assigned.
- D. The code can compile and run fine. The second line assigns a new array to list. ✔ See Figure 7.4.

See Figure 7.4.

See Figure 7.4.

The correct answer is: The code can compile and run fine. The second line assigns a new array to list.

Question 10

Correct

Mark 1.00 out of 1.00

Analyze the following code:

```
public class Test {
  public static void main(String[] args) {
    int[] x = new int[5];
    int i;
    for (i = 0; i < x.length; i++)
      x[i] = i;
    System.out.println(x[i]);
  }
}
```

Select one:

- A. The program has a runtime error because the last statement in the main method causes ArrayIndexOutOfBoundsException. ✔ After the for loop i is 5. x[5] is out of bounds.
- B. The program has a compile error because i is not defined in the last statement in the main method.
- C. The program displays 4.
- D. The program displays 0 1 2 3 4.

After the for loop i is 5. x[5] is out of bounds.

After the for loop i is 5. x[5] is out of bounds.

The correct answer is: The program has a runtime error because the last statement in the main method causes ArrayIndexOutOfBoundsException.

Question 11

Correct

Mark 2.00 out of 2.00

When you invoke a method with a parameter, the value of the argument is passed to the parameter. This is referred to as _____.

Select one:

- A. pass by reference
- B. pass by value ✔ See the third paragraph in this section.
- C. method invocation
- D. pass by name

See the third paragraph in this section.

See the third paragraph in this section.

The correct answer is: pass by value

Question 12

Correct

Mark 1.00 out of 1.00

What is the output of the following code?

```
double[] myList = {1, 5, 5, 5, 5, 1};
double max = myList[0];
int indexOfMax = 0;
for (int i = 1; i < myList.length; i++) {
if (myList[i] > max) {
  max = myList[i];
  indexOfMax = i;
 }
}
System.out.println(indexOfMax);
```

Select one:

- A. 0
- B. 1 ✔ The code displays the index of the first max value in the array.
- C. 2
- D. 3
- E. 4

The code displays the index of the first max value in the array.

The code displays the index of the first max value in the array.

The correct answer is: 1

Question 13

Correct

Mark 2.00 out of 2.00

All Java applications must have a method _____.

Select one:

- A. public static main(String[] args)
- B. public static void main(String[] args) ✔ Java application's starting method is the main method.
- C. public static Main(String args[])
- D. public static Main(String[] args)
- E. public void main(String[] args)

Java application's starting method is the main method.

Java application's starting method is the main method.

The correct answer is: public static void main(String[] args)

Question 14

Correct

Mark 1.00 out of 1.00

Analyze the following code:

```
public class Test {
  public static void main(String[] args) {
    int[] x = {1, 2, 3, 4};
    int[] y = x;

    x = new int[2];

    for (int i = 0; i < y.length; i++)
      System.out.print(y[i] + " ");
  }
}
```

Select one:

- A. The program displays 0 0 0 0
- B. The program displays 0 0 3 4
- C. The program displays 0 0
- D. The program displays 1 2 3 4 ✔ y is {1, 2, 3, 4} and x is {0, 0}.

y is {1, 2, 3, 4} and x is {0, 0}.

y is {1, 2, 3, 4} and x is {0, 0}.

The correct answer is: The program displays 1 2 3 4

Question 15

Correct

Mark 1.00 out of 1.00

Given the following program:

```
public class Test {
  public static void main(String[] args) {
    for (int i = 0; i < args.length; i++) {
      System.out.print(args[i] + " ");
    }
  }
}
```

What is the output, if you run the program using java Test 1 2 3?

Select one:

- A. 3
- B. 1
- C. 1 2
- D. 1 2 3 ✔ See Section 7.13.2.

See Section 7.13.2.

See Section 7.13.2.

The correct answer is: 1 2 3

Question 16

Correct

Mark 2.00 out of 2.00

The _____ method sorts the array scores of the double[] type.

Select one:

- A. java.util.Arrays.sorts(scores)
- B. java.util.Arrays.sortArray(scores)
- C. java.util.Arrays.sort(scores) ✔ See the second paragraph in this section.
- D. java.util.Arrays(scores)

See the second paragraph in this section.

See the second paragraph in this section.

The correct answer is: java.util.Arrays.sort(scores)

Question 17

Correct

Mark 1.00 out of 1.00

What is the output of the following code?

```
int[] myList = {1, 2, 3, 4, 5, 6};

for (int i = myList.length - 2; i >= 0; i--) {
  myList[i + 1] = myList[i];
}

for (int e: myList)
  System.out.print(e + " ");
```

Select one:

- A. 2 3 4 5 6 1
- B. 1 1 2 3 4 5 ✔ See Shifting elements in Item #8.
- C. 6 1 2 3 4 5
- D. 6 2 3 4 5 1
- E. 1 2 3 4 5 6

See Shifting elements in Item #8.

See Shifting elements in Item #8.

The correct answer is: 1 1 2 3 4 5

Question 18

Correct

Mark 2.00 out of 2.00

You should fill in the blank in the following code with _____.

```
public class Test {
  public static void main(String[] args) {
    System.out.print("The grade is ");
    printGrade(78.5);

    System.out.print("The grade is ");
    printGrade(59.5);
  }

  public static _____ printGrade(double score) {
    if (score >= 90.0) {
      System.out.println('A');
    }
    else if (score >= 80.0) {
      System.out.println('B');
    }
    else if (score >= 70.0) {
      System.out.println('C');
    }
    else if (score >= 60.0) {
      System.out.println('D');
    }
    else {
      System.out.println('F');
    }
  }
}
```

Select one:

- A. char
- B. boolean
- C. double
- D. void ✔ void should be used here because the method does not return any value.
- E. int

void should be used here because the method does not return any value.

void should be used here because the method does not return any value.

The correct answer is: void

Question 19

Incorrect

Mark 0.00 out of 2.00

Analyze the following code:

```
public class Test {
 public static void main(String[] args) {
 double[] x = {2.5, 3, 4};
 for (double value: x)
 System.out.print(value + " ");
 }
}
```

Select one:

- A. The program displays 2.5, 3, 4
- B. The program displays 2.5 3 4
- C. The program displays 2.5 3.0 4.0
- D. The program displays 2.5, 3.0 4.0
- E. The program has a syntax error because value is undefined. ✖

See the first paragraph.

The correct answer is: The program displays 2.5 3.0 4.0

Question 20

Correct

Mark 2.00 out of 2.00

Analyze the following code.

```
public class Test {
  public static void main(String[] args) {
    System.out.println(m(2));
  }

  public static int m(int num) {
    return num;
  }

  public static void m(int num) {
    System.out.println(num);
  }
}
```

Select one:

- A. The program runs and prints 2 twice.
- B. The program runs and prints 2 once.
- C. The program has a compile error because the two methods m have the same signature. ✔ You cannot overload the methods based on the type returned. See the first Note box in this section.
- D. The program has a compile error because the second m method is defined, but not invoked in the main method.

You cannot overload the methods based on the type returned. See the first Note box in this section.

You cannot overload the methods based on the type returned. See the first Note box in this section.

The correct answer is: The program has a compile error because the two methods m have the same signature.

Question 21

Incorrect

Mark 0.00 out of 1.00

Assume int[] scores = {1, 20, 30, 40, 50}, what value does java.util.Arrays.binarySearch(scores, 30) return?

Select one:

- A. 0
- B. -1 ✖
- C. 1
- D. 2
- E. -2

See the fourth paragraph in this section.

The correct answer is: 2

Question 22

Correct

Mark 2.00 out of 2.00

How can you get the word "abc" in the main method from the following call?

java Test "+" 3 "abc" 2

Select one:

- A. args[0]
- B. args[1]
- C. args[2] ✔ See Section 7.13.2.
- D. args[3]

See Section 7.13.2.

See Section 7.13.2.

The correct answer is: args[2]

Question 23

Correct

Mark 1.00 out of 1.00

Which correctly creates an array of five empty Strings?

Select one:

- A. String[] a = {"", "", "", "", ""}; ✔ See LiveExample 7.9.
- B. String[ ] a = new String [5]; for (int i = 0; i < 5; a[i++] = null);
- C. String[] a = new String [5];
- D. String[5] a;

See LiveExample 7.9.

See LiveExample 7.9.

The correct answer is: String[] a = {"", "", "", "", ""};

Question 24

Correct

Mark 1.00 out of 1.00

Analyze the following code:

```
public class Test {
  public static void main(String argv[]) {
    System.out.println("argv.length is " + argv.length);
  }
}
```

Select one:

- A. The program has a compile error because String argv[] is wrong and it should be replaced by String[] args.
- B. The program has a compile error because String argv[] is wrong and it should be replaced by String args[].
- C. If you run this program without passing any arguments, the program would display argv.length is 0. ✔ The parameter for the main method is an array of String. The declaration String argv[] is correct. When you run the program without passing arguments, argv is new String[0]. Thus, argv.length is 0. See the NOTE box in the section, "Passing Arguments to Java Programs."
- D. If you run this program without passing any arguments, the program would have a runtime error because argv is null.

The parameter for the main method is an array of String. The declaration String argv[] is correct. When you run the program without passing arguments, argv is new String[0]. Thus, argv.length is 0. See the NOTE box in the section, "Passing Arguments to Java Programs."

The parameter for the main method is an array of String. The declaration String argv[] is correct. When you run the program without passing arguments, argv is new String[0]. Thus, argv.length is 0. See the NOTE box in the section, "Passing Arguments to Java Programs."

The correct answer is: If you run this program without passing any arguments, the program would display argv.length is 0.

Question 25

Correct

Mark 1.00 out of 1.00

In the following code, what is the output for list1?

```
public class Test {
  public static void main(String[] args) {
    int[] list1 = {1, 2, 3};
    int[] list2 = {1, 2, 3};
    list2 = list1;
    list1[0] = 0; list1[1] = 1; list2[2] = 2;

    for (int i = 0; i < list1.length; i++)
      System.out.print(list1[i] + " ");
  }
}
```

Select one:

- A. 1 1 1
- B. 1 2 3
- C. 0 1 2 ✔ list2 = list1 makes list2 point to the same array as list1. See Figure 7.4.
- D. 0 1 3

list2 = list1 makes list2 point to the same array as list1. See Figure 7.4.

list2 = list1 makes list2 point to the same array as list1. See Figure 7.4.

The correct answer is: 0 1 2

Question 26

Correct

Mark 1.00 out of 1.00

Do the following two programs produce the same result?

Program I:

```
public class Test {
  public static void main(String[] args) {
    int[] list = {1, 2, 3, 4, 5};
    for (int i = 0; i < list.length; i++)
      System.out.print(list[i] + " ");
  }
}
```

Program II:

```
public class Test {
public static void main(String[] args) {
    int[] oldList = {1, 2, 3, 4, 5};
    reverse(oldList);
    for (int i = 0; i < oldList.length; i++)
      System.out.print(oldList[i] + " ");
  }

  public static void reverse(int[] list) {
    int[] newList = new int[list.length];

    for (int i = 0; i < list.length; i++)
      newList[i] = list[list.length - 1 - i];

    list = newList;
  }
}
```

Select one:

- A. Yes ✔ In Program II, the reference of newList is assigned to list. This does not change the reference for list after calling reverse. So, list is not changed after the call.
- B. No

In Program II, the reference of newList is assigned to list. This does not change the reference for list after calling reverse. So, list is not changed after the call.

In Program II, the reference of newList is assigned to list. This does not change the reference for list after calling reverse. So, list is not changed after the call.

The correct answer is: Yes

Question 27

Incorrect

Mark 0.00 out of 1.00

Analyze the following code:

```
public class Test {
  public static void main(String[] args) {
    int[] oldList = {1, 2, 3, 4, 5};
    reverse(oldList);
    for (int i = 0; i < oldList.length; i++)
      System.out.print(oldList[i] + " ");
  }

  public static void reverse(int[] list) {
    int[] newList = new int[list.length];

    for (int i = 0; i < list.length; i++)
      newList[i] = list[list.length - 1 - i];

    list = newList;
  }
}
```

Select one:

- A. The program displays 1 2 3 4 5.
- B. The program displays 1 2 3 4 5 and then raises an ArrayIndexOutOfBoundsException.
- C. The program displays 5 4 3 2 1.
- D. The program displays 5 4 3 2 1 and then raises an ArrayIndexOutOfBoundsException. ✖

The contents of the array oldList have not been changed as result of invoking the reverse method.

The correct answer is: The program displays 1 2 3 4 5.

Question 28

Correct

Mark 2.00 out of 2.00

Analyze the following code:

```
class Test {
  public static void main(String[] args) {
    System.out.println(xmethod(5));
  }

  public static int xmethod(int n, long t) {
    System.out.println("int");
    return n;
  }

  public static long xmethod(long n) {
    System.out.println("long");
    return n;
  }
}
```

Select one:

- A. The program displays int followed by 5.
- B. The program displays long followed by 5. ✔ See Listing 6.9.
- C. The program runs fine but displays things other than 5.
- D. The program does not compile because the compiler cannot distinguish which xmethod to invoke.

See Listing 6.9.

See Listing 6.9.

The correct answer is: The program displays long followed by 5.

Question 29

Correct

Mark 2.00 out of 2.00

_____ is a simple but incomplete version of a method.

Select one:

- A. A non-main method
- B. A method developed using top-down approach
- C. A main method
- D. A stub ✔ See Section 6.11.2.

See Section 6.11.2.

See Section 6.11.2.

The correct answer is: A stub

Question 30

Incorrect

Mark 0.00 out of 1.00

Analyze the following code:

```
public class Test {
  public static void main(String[] args) {
    final int[] x = {1, 2, 3, 4};
    int[] y = x;

    x = new int[2];

    for (int i = 0; i < y.length; i++)
      System.out.print(y[i] + " ");
  }
}
```

Select one:

- A. The program has a compile error on the statement x = new int[2], because x is final and cannot be changed.
- B. The program displays 1 2 3 4. ✖
- C. The elements in the array x cannot be changed, because x is final.
- D. The program displays 0 0.

The value stored in x is final, but the values in the array are not final. x is a constant reference variable that points to an array with four elements Because it is a constant, you cannot create a new reference variable x that points to a different array, but you can change the value of the elements in the array, e.g., x[1] could be changed to 10 instead of 2.

The correct answer is: The program has a compile error on the statement x = new int[2], because x is final and cannot be changed.

Question 31

Correct

Mark 2.00 out of 2.00

The signature of a method consists of _____.

Select one:

- A. parameter list
- B. method name
- C. method name and parameter list ✔ By definition, a method signature consists of a method name and parameter list.
- D. return type, method name, and parameter list

By definition, a method signature consists of a method name and parameter list.

By definition, a method signature consists of a method name and parameter list.

The correct answer is: method name and parameter list

Question 32

Incorrect

Mark 0.00 out of 2.00

What is k after the following block executes?

```
{
  int k = 2;
  nPrint("A message", k);
}
System.out.println(k);
```

Select one:

- A. 0
- B. 1
- C. 2 ✖
- D. k is not defined outside the block. So, the program has a compile error

k is defined inside the block. Outside the block, k is not defined.

The correct answer is: k is not defined outside the block. So, the program has a compile error

Question 33

Correct

Mark 1.00 out of 1.00

When you return an array from a method, the method returns _____.

Select one:

- A. a copy of the first element
- B. the length of the array
- C. a copy of the array
- D. the reference of the array ✔ See the Key Point of this section.

See the Key Point of this section.

See the Key Point of this section.

The correct answer is: the reference of the array

Question 34

Correct

Mark 1.00 out of 1.00

The _____ method copies the sourceArray to the targetArray.

Select one:

- A. System.copyArrays(sourceArray, 0, targetArray, 0, sourceArray.length);
- B. System.copyarrays(sourceArray, 0, targetArray, 0, sourceArray.length);
- C. System.arraycopy(sourceArray, 0, targetArray, 0, sourceArray.length); ✔ See the discussion on the arraycopy method in this section.
- D. System.arrayCopy(sourceArray, 0, targetArray, 0, sourceArray.length);

See the discussion on the arraycopy method in this section.

See the discussion on the arraycopy method in this section.

The correct answer is: System.arraycopy(sourceArray, 0, targetArray, 0, sourceArray.length);

Question 35

Correct

Mark 1.00 out of 1.00

Analyze the following code.

```
public class Test {
  public static void main(String[] args) {
    int[] x = new int[3];
    System.out.println("x[0] is " + x[0]);
  }
}
```

Select one:

- A. The program has a compile error because the size of the array wasn't specified when declaring the array.
- B. The program has a runtime error because the array element x[0] is not defined.
- C. The program has a runtime error because the array elements are not initialized.
- D. The program runs fine and displays x[0] is 0. ✔ The default value for an int array element is 0.

The default value for an int array element is 0.

The default value for an int array element is 0.

The correct answer is: The program runs fine and displays x[0] is 0.

Question 36

Incorrect

Mark 0.00 out of 1.00

Analyze the following code:

```
public class Test {
  public static void main(String[] args) {
    int[] x = {1, 2, 3, 4};
    int[] y = x;

    x = new int[2];

    for (int i = 0; i < x.length; i++)
      System.out.print(x[i] + " ");
  }
}
```

Select one:

- A. The program displays 0 0 3 4 ✖
- B. The program displays 0 0 0 0
- C. The program displays 1 2 3 4
- D. The program displays 0 0

y is {1, 2, 3, 4} and x is {0, 0}

The correct answer is: The program displays 0 0

Question 37

Correct

Mark 2.00 out of 2.00

(char)('a' + Math.random() * ('z' - 'a' + 1)) returns a random character _____.

Select one:

- A. between 'b' and 'y'
- B. between 'a' and 'z' ✔ See LiveExample 6.10.
- C. between 'b' and 'z'
- D. between 'a' and 'y'

See LiveExample 6.10.

See LiveExample 6.10.

The correct answer is: between 'a' and 'z'

Question 38

Correct

Mark 1.00 out of 1.00

When you pass an array to a method, the method receives _____.

Select one:

- A. the reference of the array ✔ See the code animation for TestArrayArguments.
- B. a copy of the array
- C. the length of the array
- D. a copy of the first element

See the code animation for TestArrayArguments.

See the code animation for TestArrayArguments.

The correct answer is: the reference of the array

Question 39

Correct

Mark 1.00 out of 1.00

Suppose int i = 5, which of the following can be used as an index for array double[] t = new double[100]? Please select all that apply.

Select one or more:

- A. i ✔ The array t has 100 elements and its index is an integer from 0 to 99. So, A, B, C are all correct. D and E are double values, not integers.
- B. (int)(Math.random() * 100)) ✔ The array t has 100 elements and its index is an integer from 0 to 99. So, A, B, C are all correct. D and E are double values, not integers.
- C. i + 10 ✔ The array t has 100 elements and its index is an integer from 0 to 99. So, A, B, C are all correct. D and E are double values, not integers.
- D. i + 6.5
- E. Math.random() * 100

The array t has 100 elements and its index is an integer from 0 to 99. So, A, B, C are all correct. D and E are double values, not integers.

The array t has 100 elements and its index is an integer from 0 to 99. So, A, B, C are all correct. D and E are double values, not integers.

The correct answers are: i, (int)(Math.random() * 100)), i + 10

Question 40

Correct

Mark 2.00 out of 2.00

The client can use a method without knowing how it is implemented. The details of the implementation are encapsulated in the method and hidden from the client who invokes the method. This is known as _____. Please select all that apply.

Select one or more:

- A. encapsulation ✔ See the first paragraph.
- B. simplifying method
- C. method hiding
- D. information hiding ✔ See the first paragraph.

See the first paragraph.

See the first paragraph.

The correct answers are: information hiding, encapsulation

Question 41

Correct

Mark 1.00 out of 1.00

What is output of the following code?

```
public class Test {
  public static void main(String[] args) {
    int list[] = {1, 2, 3, 4, 5, 6};

    for (int i = 1; i < list.length; i++)
      list[i] = list[i - 1];

    for (int i = 0; i < list.length; i++)
      System.out.print(list[i] + " ");
  }
}
```

Select one:

- A. 1 1 1 1 1 1 ✔ The first element is copied to the second, the second is then copied to the third, and so on.
- B. 2 3 4 5 6 1
- C. 1 2 3 4 5 6
- D. 2 3 4 5 6 6

The first element is copied to the second, the second is then copied to the third, and so on.

The first element is copied to the second, the second is then copied to the third, and so on.

The correct answer is: 1 1 1 1 1 1

Question 42

Correct

Mark 1.00 out of 1.00

Which code fragment would correctly identify the number of arguments passed via the command line to a Java application, excluding the name of the class that is being invoked?

Select one:

- A. int count = args.length - 1;
- B. int count=0; while (!(args[count].equals(""))) count ++;
- C. int count = args.length; ✔ See LiveExample 7.9.
- D. int count = 0; while (args[count] != null) count ++;

See LiveExample 7.9.

See LiveExample 7.9.

The correct answer is: int count = args.length;

Question 43

Correct

Mark 2.00 out of 2.00

Analyze the following code:

```
public class Test {
  public static void main(String[] args) {
    System.out.println(xMethod(5, 500L));
  }

  public static int xMethod(int n, long l) {
    System.out.println("int, long");
    return n;
  }

  public static long xMethod(long n, long l) {
    System.out.println("long, long");
    return n;
  }
}
```

Select one:

- A. The program displays int, long followed by 5. ✔ See Listing 6.9.
- B. The program displays long, long followed by 5.
- C. The program runs fine but displays things other than 5.
- D. The program does not compile because the compiler cannot distinguish which xmethod to invoke.

See Listing 6.9.

See Listing 6.9.

The correct answer is: The program displays int, long followed by 5.

Question 44

Incorrect

Mark 0.00 out of 1.00

The JVM stores the array in an area of memory, called _____, which is used for dynamic memory allocation where blocks of memory are allocated and freed in an arbitrary order.

Select one:

- A. memory block
- B. stack ✖
- C. dynamic memory
- D. heap

See the Note after Figure 5.7.

The correct answer is: heap

Question 45

Correct

Mark 1.00 out of 1.00

Analyze the following code:

```
public class Test {
  public static void main(String[] args) {
    int[] a = new int[4];
    a[1] = 1;
    a = new int[2];
    System.out.println("a[1] is " + a[1]);
  }
}
```

Select one:

- A. The program displays a[1] is 1.
- B. The program has a runtime error because a[1] is not initialized.
- C. The program displays a[1] is 0. ✔ After executing the statement a = new int[2], a refers to int[2]. The default value for a[0] and a[1] is 0.
- D. The program has a compile error because new int[2] is assigned to a.

After executing the statement a = new int[2], a refers to int[2]. The default value for a[0] and a[1] is 0.

After executing the statement a = new int[2], a refers to int[2]. The default value for a[0] and a[1] is 0.

The correct answer is: The program displays a[1] is 0.

Question 46

Correct

Mark 2.00 out of 2.00

_____ is to implement one method in the structure chart at a time from the top to the bottom.

Select one:

- A. Stepwise refinement
- B. Bottom-up and top-down approach
- C. Top-down approach ✔ See Section 6.11.2.
- D. Bottom-up approach

See Section 6.11.2.

See Section 6.11.2.

The correct answer is: Top-down approach

Question 47

Correct

Mark 2.00 out of 2.00

Show the output of the following code:

```
public class Test {
  public static void main(String[] args) {
    int[] x = {1, 2, 3, 4, 5};
    increase(x);

    int[] y = {1, 2, 3, 4, 5};
    increase(y[0]);

    System.out.println(x[0] + " " + y[0]);
  }

  public static void increase(int[] x) {
    for (int i = 0; i < x.length; i++)
      x[i]++;
  }

  public static void increase(int y) {
    y++;
  }
}
```

Select one:

- A. 0 0
- B. 1 1
- C. 2 1 ✔ Invoking increase(x) passes the reference of the array to the method. Invoking increase(y[0]) passes the value 1 to the method. The value y[0] outside the method is not changed.
- D. 1 2
- E. 2 2

Invoking increase(x) passes the reference of the array to the method. Invoking increase(y[0]) passes the value 1 to the method. The value y[0] outside the method is not changed.

Invoking increase(x) passes the reference of the array to the method. Invoking increase(y[0]) passes the value 1 to the method. The value y[0] outside the method is not changed.

The correct answer is: 2 1

Question 48

Correct

Mark 1.00 out of 1.00

The reverse method is defined in the textbook. What is list1 after executing the following statements?

```
int[] list1 = {1, 2, 3, 4, 5, 6};
list1 = reverse(list1);
```

Select one:

- A. list1 is 6 6 6 6 6 6
- B. list1 is 1 2 3 4 5 6
- C. list1 is 0 0 0 0 0 0
- D. list1 is 6 5 4 3 2 1 ✔ The reverse method in this section returns a new array that is the reversal of the source.

The reverse method in this section returns a new array that is the reversal of the source.

The reverse method in this section returns a new array that is the reversal of the source.

The correct answer is: list1 is 6 5 4 3 2 1

Question 49

Partially correct

Mark 0.50 out of 1.00

Which of the following statements are valid? Please select all that apply.

Select one or more:

- A. int i = new int(30);
- B. double d[] = new double[30];
- C. int[] i = {3, 4, 3, 2}; ✔ E would be corrected if it is char[] c = new char[]{'a', 'b', 'c', 'd'};
- D. char[] c = new char();
- E. char[] c = new char[4]{'a', 'b', 'c', 'd'}; ✖

E would be corrected if it is char[] c = new char[]{'a', 'b', 'c', 'd'};

The correct answers are: double d[] = new double[30];, int[] i = {3, 4, 3, 2};

Question 50

Correct

Mark 2.00 out of 2.00

Does the return statement in the following method cause compile errors?

```
public static void main(String[] args) {
  int max = 0;
  if (max != 0)
    System.out.println(max);
  else
    return;
}
```

Select one:

- A. Yes
- B. No ✔ See the Caution box. It is rare, but sometimes useful to have a return statement for circumventing the normal flow of control in a void method.

See the Caution box. It is rare, but sometimes useful to have a return statement for circumventing the normal flow of control in a void method.

See the Caution box. It is rare, but sometimes useful to have a return statement for circumventing the normal flow of control in a void method.

The correct answer is: No

Question 51

Correct

Mark 1.00 out of 1.00

The reverse method is defined in this section. What is list1 after executing the following statements?

```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);
```

Select one:

- A. list1 is 1 2 3 4 5 6 ✔ See the reverse method in this section.
- B. list1 is 0 0 0 0 0 0
- C. list1 is 6 6 6 6 6 6
- D. list1 is 6 5 4 3 2 1

See the reverse method in this section.

See the reverse method in this section.

The correct answer is: list1 is 1 2 3 4 5 6

Question 52

Correct

Mark 1.00 out of 1.00

If you declare an array double[] list = new double[5], the highest index in array list is _____.

Select one:

- A. 0
- B. 1
- C. 2
- D. 3
- E. 4 ✔ See Figure 7.1.

See Figure 7.1.

See Figure 7.1.

The correct answer is: 4

Question 53

Correct

Mark 1.00 out of 1.00

Assume int[] scores = {1, 20, 30, 40, 50}, what is the output of System.out.println(java.util.Arrays.toString(scores))?

Select one:

- A. {1, 20, 30, 40, 50}
- B. [1, 20, 30, 40, 50] ✔ See the toString method in this section.
- C. {1 20 30 40 50}
- D. [1 20 30 40 50]

See the toString method in this section.

See the toString method in this section.

The correct answer is: [1, 20, 30, 40, 50]

Question 54

Correct

Mark 2.00 out of 2.00

Which of the following should be defined as a void method?

Select one:

- A. Write a method that returns a random integer from 1 to 100.
- B. Write a method that checks whether a number is from 1 to 100.
- C. Write a method that prints integers from 1 to 100. ✔ A method that does not return a value should be defined void.
- D. Write a method that converts an uppercase letter to lowercase.

A method that does not return a value should be defined void.

A method that does not return a value should be defined void.

The correct answer is: Write a method that prints integers from 1 to 100.

Question 55

Correct

Mark 1.00 out of 1.00

Which of the following statements are correct to invoke the printMax method in Listing 7.5 in the textbook? Please select all that apply.

Select one or more:

- A. printMax(1, 2, 2, 1, 4); ✔ The last one printMax(new int[]{1, 2, 3}); is incorrect, because the array must of the double[] type.
- B. printMax(new double[]{1, 2, 3}); ✔ The last one printMax(new int[]{1, 2, 3}); is incorrect, because the array must of the double[] type.
- C. printMax(1.0, 2.0, 2.0, 1.0, 4.0); ✔ The last one printMax(new int[]{1, 2, 3}); is incorrect, because the array must of the double[] type.
- D. printMax(new int[]{1, 2, 3});

The last one printMax(new int[]{1, 2, 3}); is incorrect, because the array must of the double[] type.

The last one printMax(new int[]{1, 2, 3}); is incorrect, because the array must of the double[] type.

The correct answers are: printMax(1, 2, 2, 1, 4);, printMax(new double[]{1, 2, 3});, printMax(1.0, 2.0, 2.0, 1.0, 4.0);

Question 56

Correct

Mark 1.00 out of 1.00

What is the correct term for numbers[99]?

Select one:

- A. index variable
- B. indexed variable ✔ See Section 7.2.4.
- C. array variable
- D. index
- E. array

See Section 7.2.4.

See Section 7.2.4.

The correct answer is: indexed variable

Question 57

Correct

Mark 1.00 out of 1.00

Which of the following are correct to declare an array of int values? Please select all that apply.

Select one or more:

- A. int[] a; ✔ See the first paragraph.
- B. int a;
- C. double[] a;
- D. int a[]; ✔ See the first paragraph.

See the first paragraph.

See the first paragraph.

The correct answers are: int[] a;, int a[];

Question 58

Correct

Mark 1.00 out of 1.00

Analyze the following code:

```
public class Test1 {
  public static void main(String[] args) {
    xMethod(new double[]{3, 3});
    xMethod(new double[5]);
    xMethod(new double[3]{1, 2, 3});
  }

  public static void xMethod(double[] a) {
    System.out.println(a.length);
  }
}
```

Select one:

- A. The program has a runtime error because a is null.
- B. The program has a compile error because xMethod(new double[5]) is incorrect.
- C. The program has a compile error because xMethod(new double[]{3, 3}) is incorrect.
- D. The program has a compile error because xMethod(new double[3]{1, 2, 3}) is incorrect. ✔ new double[3]{1, 2, 3} should be replaced by new double[]{1, 2, 3}) (anonymous array).

new double[3]{1, 2, 3} should be replaced by new double[]{1, 2, 3}) (anonymous array).

new double[3]{1, 2, 3} should be replaced by new double[]{1, 2, 3}) (anonymous array).

The correct answer is: The program has a compile error because xMethod(new double[3]{1, 2, 3}) is incorrect.

Question 59

Incorrect

Mark 0.00 out of 2.00

Once an array is created, its size _____.

Select one:

- A. can be changed ✖
- B. is fixed
- C. is not determined

See the Key Point.

The correct answer is: is fixed

Question 60

Correct

Mark 1.00 out of 1.00

Which of the following are incorrect? Please select all that apply.

Select one or more:

- A. int[] a = new int[2];
- B. int[] a = new int(2); ✔ See Sections 7.2.1 and 7.2.2.
- C. int a() = new int[2]; ✔ See Sections 7.2.1 and 7.2.2.
- D. int a[] = new int[2];
- E. int a = new int[2]; ✔ See Sections 7.2.1 and 7.2.2.

See Sections 7.2.1 and 7.2.2.

See Sections 7.2.1 and 7.2.2.

The correct answers are: int[] a = new int(2);, int a = new int[2];, int a() = new int[2];

Question 61

Correct

Mark 2.00 out of 2.00

You should fill in the blank in the following code with _____.

```
public class Test {
  public static void main(String[] args) {
    System.out.print("The grade is " + getGrade(78.5));
    System.out.print("\nThe grade is " + getGrade(59.5));
  }

  public static _____ getGrade(double score) {
    if (score >= 90.0)
      return 'A';
    else if (score >= 80.0)
      return 'B';
    else if (score >= 70.0)
      return 'C';
    else if (score >= 60.0)
      return 'D';
    else
      return 'F';
  }
}
```

Select one:

- A. void
- B. double
- C. boolean
- D. int
- E. char ✔ char should be placed here because the method returns a character.

char should be placed here because the method returns a character.

char should be placed here because the method returns a character.

The correct answer is: char

Question 62

Correct

Mark 1.00 out of 1.00

Which of the following is the correct header of the main method? Please select all that apply.

Select one or more:

- A. public static void main(String[] args) ✔ E is incorrect because the main method must be public.
- B. public static void main(String args[]) ✔ E is incorrect because the main method must be public.
- C. public static void main(String[] x) ✔ E is incorrect because the main method must be public.
- D. public static void main(String x[]) ✔ E is incorrect because the main method must be public.
- E. static void main(String[] args)

E is incorrect because the main method must be public.

E is incorrect because the main method must be public.

The correct answers are: public static void main(String[] args), public static void main(String args[]), public static void main(String[] x), public static void main(String x[])

Question 63

Correct

Mark 2.00 out of 2.00

(int)('a' + Math.random() * ('z' - 'a' + 1)) returns a random number _____.

Select one:

- A. between 0 and (int)'z'
- B. between (int)'a' and (int)'z' ✔ See LiveExample 6.10.
- C. between 'a' and 'y'
- D. between 'a' and 'z'

See LiveExample 6.10.

See LiveExample 6.10.

The correct answer is: between (int)'a' and (int)'z'

Question 64

Incorrect

Mark 0.00 out of 1.00

What is output of the following code?

```
public class Test {
  public static void main(String[] args) {
    int[] x = {120, 200, 16};
    for (int i = x.length - 1; i >= 0; i--)
      System.out.print(x[i] + " ");
  }
}
```

Select one:

- A. 120 16 200
- B. 200 16 120
- C. 120 200 16 ✖
- D. 16 200 120

The code displays the array elements from the last one down to the first one.

The correct answer is: 16 200 120

Question 65

Correct

Mark 2.00 out of 2.00

Given the following method

```
static void nPrint(String message, int n) {
  while (n > 0) {
    System.out.print(message);
    n--;
  }
}
```

What is k after invoking nPrint("A message", k)?

```
int k = 2;
nPrint("A message", k);
```

Select one:

- A. 0
- B. 1
- C. 2 ✔ See Listing 6.4.
- D. 3

See Listing 6.4.

See Listing 6.4.

The correct answer is: 2

Question 66

Correct

Mark 1.00 out of 1.00

Suppose a method p has the following header:

public static int[] p()

What return statement may be used in p()?

Select one:

- A. return int[]{1, 2, 3};
- B. return {1, 2, 3};
- C. return 1;
- D. return new int[]{1, 2, 3}; ✔ The correct syntax to create an array is new int[]{1, 2, 3}.

The correct syntax to create an array is new int[]{1, 2, 3}.

The correct syntax to create an array is new int[]{1, 2, 3}.

The correct answer is: return new int[]{1, 2, 3};

Question 67

Correct

Mark 2.00 out of 2.00

(int)(Math.random() * (65535 + 1)) returns a random number _____.

Select one:

- A. between 1 and 65536
- B. between 1 and 65535
- C. between 0 and 65536
- D. between 0 and 65535 ✔ See the second paragraph in this section.

See the second paragraph in this section.

See the second paragraph in this section.

The correct answer is: between 0 and 65535

Question 68

Correct

Mark 2.00 out of 2.00

Arguments to methods always appear within _____.

Select one:

- A. parentheses ✔ See Listing 6.1.
- B. quotation marks
- C. brackets
- D. curly braces

See Listing 6.1.

See Listing 6.1.

The correct answer is: parentheses

Question 69

Correct

Mark 2.00 out of 2.00

Analyze the following code:

```
public class Test {
 public static void main(String[] args) {
 System.out.println(max(1, 2));
 }

 public static double max(int num1, double num2) {
 System.out.println("max(int, double) is invoked");

 if (num1 > num2)
 return num1;
 else
 return num2;
 }

 public static double max(double num1, int num2) {
 System.out.println("max(double, int) is invoked");

 if (num1 > num2)
 return num1;
 else
 return num2;
 }
}
```

Select one:

- A. The program cannot compile because the compiler cannot determine which max method should be invoked. ✔ This is known as ambiguous method invocation. See the last Note box in this section.
- B. The program runs and prints "max(int, double) is invoked" followed by 2.
- C. The program runs and prints 2 followed by "max(int, double)" is invoked.
- D. The program cannot compile because you cannot have the print statement in a non-void method.
- E. The program runs and prints 2 followed by "max(double, int)" is invoked.

This is known as ambiguous method invocation. See the last Note box in this section.

This is known as ambiguous method invocation. See the last Note box in this section.

The correct answer is: The program cannot compile because the compiler cannot determine which max method should be invoked.

Question 70

Correct

Mark 2.00 out of 2.00

Given the following method, what is the output of the call nPrint('a', 4)?

```
static void nPrint(String message, int n) {
  while (n > 0) {
    System.out.print(message);
    n--;
  }
}
```

Select one:

- A. aaaaa
- B. aaaa
- C. aaa
- D. invalid call ✔ Invalid call because char 'a' cannot be passed to string message

Invalid call because char 'a' cannot be passed to string message

Invalid call because char 'a' cannot be passed to string message

The correct answer is: invalid call